NanoEls H5 — Wiring schematic

Connectors, pinout and signals with the Italian silkscreen names (ELS controller, firmware H5V65)

Italian rear label
Pins 1…4 = DB2EVC-3.5-4P connector
A4 sheet — print at 100%

Pin 1 is +5V, on the side of the 5V marking on the connector. The **PCB has two faces, both silkscreened**: on the face with **`5V ENA DIR STEP`** · **`5V GND B A`** · **`5V GND CLK DATA`**, +5V is the first and the pins go **1 → 4**; on the other face the markings are **`STEP DIR ENA 5V`** · **`A B GND 5V`** · **`KEYD KEYC GND 5V`** and +5V is the last, so the pins go **4 → 1**. Follow the label printed on the face you are looking at: nothing has to be inverted by hand.

AXIS OUTPUTS — Z, X, Y (STEPPER DRIVERS)

Pin	Signal	Z — GPIO	X — GPIO	Y — GPIO	Notes
1	**+5V**	—	—	—	Powers the driver optocouplers
2	**ABIL.**	41	16	1	Driver enable (ENABLE)
3	**SENSO**	42	15	2	Direction of rotation (DIRECTION)
4	**PASSO**	35	7	17	Step pulses (STEP)

Toward the driver the signals come out at **5 V** (SN74HCT245 converter). Motor power (VMOT) does not pass through here: it goes to the driver terminals.
Order: **`+5V`** · **`ABIL.`** · **`SENSO`** · **`PASSO`** = pins **1 → 4** (face with **`5V ENA DIR STEP`**); on the other face **`PASSO`** · **`SENSO`** · **`ABIL.`** · **`+5V`** = pins **4 → 1** (face with **`STEP DIR ENA 5V`**).

ENCODER AND SCALE INPUTS — MANUAL PULSES, SCALES, SPINDLE

Connector	Use	Pin 1 · +5V	Pin 2 · MASSA	Pin 3 · B	Pin 4 · A
`IMPULSI Z`	Z axis manual handwheel	+5V	MASSA	8	18
`IMPULSI X`	X axis manual handwheel	+5V	MASSA	21	47
`IMPULSI Y`	Y axis manual handwheel	+5V	MASSA	3	46
`SCALA Z`	Z axis linear scale (not read by the firmware)	+5V	MASSA	12	9
`SCALA X`	X axis linear scale (not read by the firmware)	+5V	MASSA	39	40
`SCALA Y`	Y axis linear scale (not read by the firmware)	+5V	MASSA	48	45
`ENCODER`	Spindle optical encoder	+5V	MASSA	14	13

Order: **`+5V`** · **`MASSA`** · **`B`** · **`A`** = pins **1 → 4** (face with **`5V GND B A`**); on the other face **`A`** · **`B`** · **`MASSA`** · **`+5V`** = pins **4 → 1** (face with **`A B GND 5V`**). The same for IMPULSI, SCALA and ENCODER.

JOYSTICK — SEGN. JOY AND ALIM. JOY

Pin	Signal	GPIO
1	**`JB`**	38
2	**`JY`**	6
3	**`JX`**	5
4	**`JZ`**	4

SEGN. JOY: three analog axes + button. Pin 1 = JB (on the face with **`JZ JX JY JBUT`** it is JBUT).

Pin	Signal
1	**+5V**
2	**+5V**
3	**MASSA**
4	**MASSA**

ALIM. JOY: joystick power supply (double contact). Pin 1 = +5V.

KEYBOARD, POWER SUPPLY, FREE I/O AND DISPLAY

Pin	TASTIERA (PS/2)	GPIO
1	**+5V**	—
2	**MASSA**	—
3	**CLK**	36
4	**DATI**	37

TASTIERA: PS/2 keyboard (clock/data). Pin 1 = +5V.

Pin	ALIM. 5V	GPIO
1	**+5V**	—
2	**+5V**	—
3	**MASSA**	—
4	**MASSA**	—

ALIM. 5V: board power input. Pin 1 = +5V.

Pin	VARIE	GPIO
1	**+5V**	—
2	**MASSA**	—
3	**IO10**	10
4	**IO11**	11

VARIE: two free I/O (relay, probe, auxiliary signals). Pin 1 = +5V.

Pin	UART (Nextion display)	GPIO
1	**+5V**	—
2	**MASSA**	—
3	**RX**	44
4	**TX**	43

UART: pin 1 = +5V, pin 4 = TX. Separate connector next to the NanoEls H5 marking, on the short side opposite the ESP32-S3 module; not on the rear label (UART0, shared with the USB monitor).

- **Board silkscreen**: **`ZPULSE / XPULSE / YPULSE`** = IMPULSI Z/X/Y · **`ZSCALE / XSCALE / YSCALE`** = SCALA Z/X/Y · **`JSIGNAL`** = SEGN. JOY · **`JPOWER`** = ALIM. JOY · **`POWER`** = ALIM. 5V · **`MISC`** = VARIE · **`KEYBOARD`** = TASTIERA · **`STEP / DIR / ENA`** = PASSO / SENSO / ABIL. · **`GND`** = MASSA · **`DATA`** = DATI · **`JBUTTON`** = JB.
- All inputs are **5 V tolerant**; the axis outputs already come out at 5 V. The Nextion display connects to the 4-pin UART connector.
- Default values: handwheels **600 pulses/rev**, spindle encoder **1200 pulses/rev**. They are changed from the web UI.
- The joystick is **off by default**: enable it in SETUP if it is used.
- The linear scales **SCALA Z / SCALA X / SCALA Y** are wired on the board but firmware H5V65 does not read them.
- The rear label lists the signals in the order **`5V`** · **`ABIL.`** · **`SENSO`** · **`PASSO`** (that is, from pin 1 to pin 4) and **`5V`** · **`MASSA`** · **`B`** · **`A`** for the inputs.

Before connecting: ELS off, check that pin 1 is +5V (the side of the 5V marking) and that the driver STEP/DIRECTION/ENABLE inputs are optically isolated. Do not bring motor voltage to the board connectors.